PCSPF

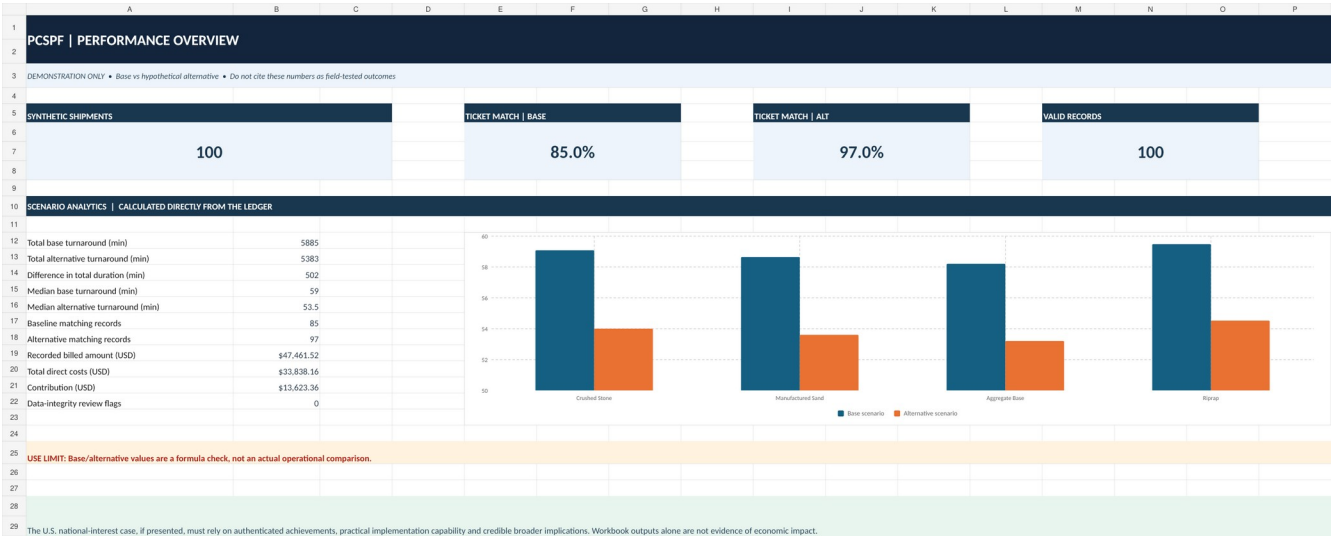
TECHNICAL VALIDATION REPORT

Port-Corridor SME Performance Framework

Operations Toolkit v1.1 | Microsoft Excel | U.S. market conventions

Deterministic logic • Ledger reconciliation • Evidence limits • Field-readiness controls

Illustrative performance interface — actual workbook preview



VERIFICATION SCOPE / NO EXTERNAL ATTESTATION

The conclusions address the supplied, unchanged 100-record synthetic workbook and selected spreadsheet controls. This is an internal analytical verification, not an independent expert opinion, a real U.S. pilot, a financial audit, or a certification of economic impact.

Release date: November 9, 2026 | Project owner: Ricardo Lopes Wiest

Companion files: PCSPF_v1_1_US_Operations_Toolkit_QA_Corrected.xlsx; PCSPF_v1_0_User_and_Implementation_Guide_EN.docx

01 Executive verification conclusion

OUTCOME • PASS, BOUNDED TO THE ORIGINAL SYNTHETIC SAMPLE

All 13 built-in reference checks report PASS on the unmodified workbook. An independent recomputation of the prescribed sample and of the main ledger rollups confirms those reference values. This conclusion does not extend to production reliability, real-world business results or national-scale use.

Disposition	Finding	Interpretation
PASS	13 of 13 workbook QA reference checks	Original 100-record deterministic data only.
PASS	Independent computational reproduction	Shipment counts, constructed times, ticket matches, cost totals and product summaries.
PASS	Internal arithmetic reconciliation	Billing = recorded direct costs + calculated contribution, within monetary rounding.
LIMITED	Direct source-data guardrails	Selected negative numeric values raised row-level CHECK; comprehensive robustness is not established.
PASS (INTERNAL)	Dynamic QA propagation following adverse edits	Four controlled edits triggered expected row-level and dependent QA/dashboard changes; separately confirmed after .xlsx reimport. Native Excel and external verification are still pending.
NOT TESTED	Field effectiveness / independent adoption	No real company data, operating trial, independent technical opinion, or U.S. adopter.

Verified reference values

Measure	Result	Meaning
Synthetic shipments	100	Four commodities; 25 each.
Ticket reference matches	85 base / 97 alternative	Constructed scenarios, not a recorded field change.
Turnaround totals	5,885 / 5,383 minutes	502-minute programmed difference; not savings.
Turnaround medians	59.0 / 53.5 minutes	Medians not additive.
Billed amount / direct costs	US$ 47,461.52 / US$ 33,838.16	Amounts are fictional.
Direct contribution	US$ 13,623.36	Not profit, EBITDA, or cash generation.

02 Review object, artifacts and provenance

The review concerns the corrected v1.1 workbook issued November 9, 2026. The review used its actual formulas and calculated values, the QA TESTS sheet, independent reconstruction of the 100 synthetic transactions, and controlled adverse-input mutation tests. No third-party company supplied operational data or results.

Property	Verified record
Primary artifact	PCSPF_v1_1_US_Operations_Toolkit_QA_Corrected.xlsx
Analytical coverage	100 synthetic loads; 12 synthetic equipment-period records.
Reference as-of date	November 9, 2026; CONTROL!B7.
Workbook worksheets	12 named worksheets, including TRANSACTIONS, EQUIPMENT LOG, DASHBOARD and QA TESTS.
Transaction input / output	TRANSACTIONS!A7:V206 / W7:AL206; rows 7–106 populated.
Equipment input / output	EQUIPMENT LOG!A7:K46 / L7:R46; rows 7–18 populated.
Baseline specification	Professional Plan, Chapters 4, 15, 16 and 22–23.
Review purpose	Demonstrate selected arithmetic and control behavior; identify limitations for external review.

File identity and reproducibility

SHA-256 (workbook byte-for-byte reference):

687790b8141c21f635585bcf2278055ab9ae4a8d17bd8689616d0e04d23c7caf

The checksum binds this report to the corrected v1.1 workbook release, not to later edited copies. Any change to the file requires a new checksum, revision record and fresh QA results.

AUDIT AUTHORSHIP AND ASSURANCE

This document records an internal technical assessment. It is not signed by an independent U.S. expert, a certified public accountant, an engineering specialist or a law firm. Any third-party reviewer must conduct and document their own tests.

03 Methodology and test boundaries

Review procedure

Step	Procedure	Evidence
1	Import supplied .xlsx; inspect source input ranges and formula-driven summary cells.	Workbook read-back.
2	Read 13 fixed expected values and statuses from QA TESTS!A7:D19.	Reference acceptance matrix.
3	Recompute synthetic durations and invoice-reference matches independently from the specified arithmetic recipe.	100-record Python verification.
4	Independently sum billing, direct costs, contribution and commodity groupings from ledger inputs.	Cross-tab reconciliation.
5	Inspect receivable classifications and independent equipment-ledger count.	A/R and fleet controls.
6	Apply four negative/adverse changes in a temporary in-memory copy; inspect both row-level and dependent dashboard/QA values, then restore original data and verify clean result.	Mutation audit: duplicate ID; invoice-ticket mismatch; negative quantity; negative equipment fuel.

Acceptance criteria

- The unchanged sample must return all 13 programmed expected values; each formula result must display PASS.
- Independent recomputation must agree with sample definitions and summary amounts to practical rounding precision.
- No field effects, adoption or financial savings may be inferred from a constructed alternative scenario.
- Any failed, untested or non-reproduced feature must be reported explicitly rather than presumed operational.

CRITICAL DISTINCTION

A successful computational smoke test demonstrates that a specific sample is internally coherent. It does not establish that the same workbook safely handles messy client data or that a control intervention improves a real company.

04 Reference acceptance matrix: 13 checks

Check	Observed	Expected	Result
Sample shipment count	100	100	PASS
Baseline matched tickets	85	85	PASS
Alternative matched tickets	97	97	PASS
Base duration sum, min	5,885	5,885	PASS
Alternative duration sum, min	5,383	5,383	PASS
Median base turnaround, min	59	59	PASS
Median alternative turnaround, min	53.5	53.5	PASS
Total base/alternative difference, min	502	502	PASS
Duplicate shipment IDs	0	0	PASS
Transaction integrity exceptions	0	0	PASS
Equipment integrity exceptions	0	0	PASS
Total loads across products	100	100	PASS
Accounts receivable count	100	100	PASS

ACCEPTANCE DECISION

13 / 13 checks passed for the fixed synthetic reference. Acceptance tests verify arithmetic and selected invariants only. The expected-value table must be redesigned before loading real-company data.

Trace locations

Reference checks: QA TESTS!A7:E19; aggregate gate: QA TESTS!A22:D22. The source cohort occupies TRANSACTIONS rows 7–106 and EQUIPMENT LOG rows 7–18. Check names and results reproduce the worksheet, not an externally certified test report.

05 Deterministic reproduction and data logic

The Professional Plan specifies integer IDs SYN-001 through SYN-100 with deterministic arithmetic. For i = 1..100, the synthetic baseline duration is 40 + 4 × (i mod 9) + (i mod 7); the alternative is baseline − ((7i) mod 11). Baseline matching holds when i mod 13 is neither 0 nor 1; alternative matching holds when i mod 31 is not 0.

Independent computation	Reference result	Verification
Sample indices	1–100	100 unique source IDs.
Matching invoice/ticket references	85 vs. 97	Derived from exact reference logic.
Summed scenario minutes	5,885 vs. 5,383	Summation of all 100 durations.
Medians	59.0 vs. 53.5 minutes	Computed on complete distributions.
Programmed difference	502 minutes	5,885 − 5,383.
Unresolved core input errors in sample	0	No displayed CHECK in populated synthetic rows.

Selected workbook formulas

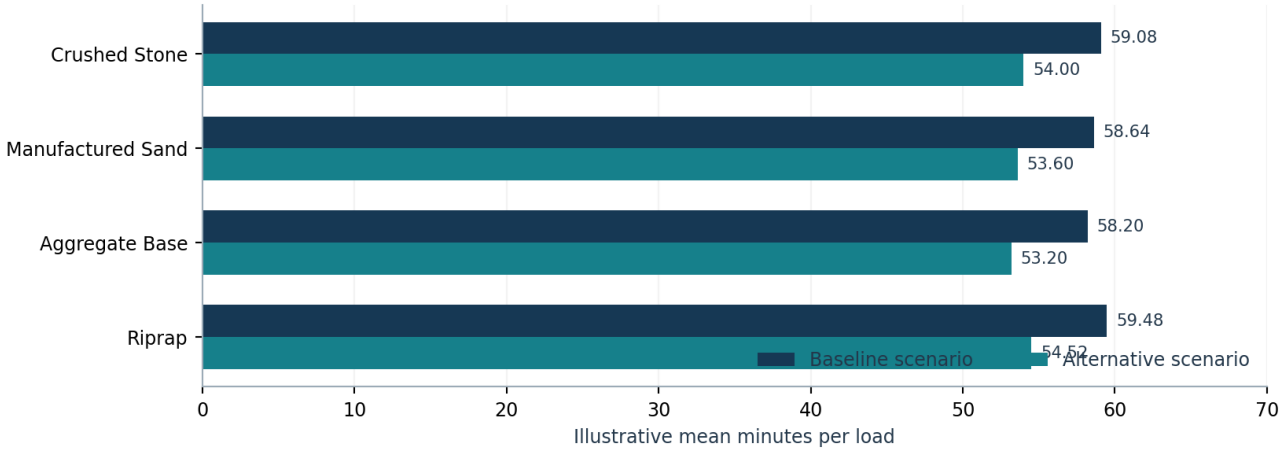
Target field	Core rule / restriction
TRANSACTIONS W/X	1 if the source load-ticket identifier exactly matches the scenario invoice reference; else 0.
TRANSACTIONS Y/Z	Rounded minutes = (departure timestamp − arrival timestamp) × 1,440.
TRANSACTIONS AL	Base duration − alternative duration, computed per row.
DASHBOARD B12:B18	Sums, medians and match counts read from calculated transaction rows.
TRANSACTIONS AK	Selected missingness, nonnegative amount, positive-hour and timestamp-order rules.

INTERPRETATION GUARDRAIL

The alternative dataset was generated to differ from baseline. The 502-minute difference and the 12-point match-rate increase are not experimentally measured improvements and may not be represented as business savings.

06 Dispatch analytics

All four commodity groups contain 25 synthetic loads. The chart reproduces the baseline and alternative average turnaround from DISPATCH ANALYSIS!A7:H12. It illustrates summary logic and is not a field performance comparison.

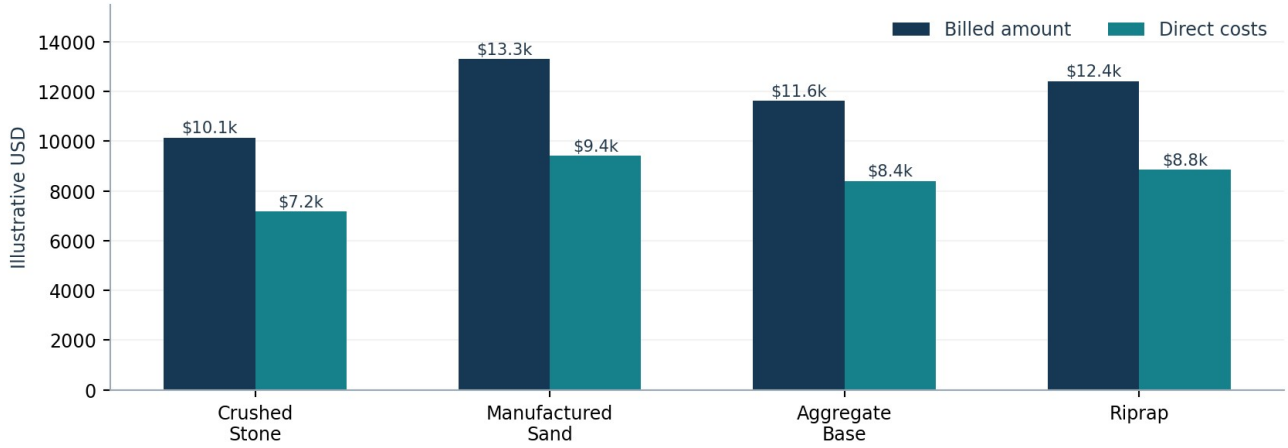


Product	Base total	Alt total	Constructed delta
Crushed Stone	1,477 min	1,350 min	127 min
Manufactured Sand	1,466 min	1,340 min	126 min
Aggregate Base	1,455 min	1,330 min	125 min
Riprap	1,487 min	1,363 min	124 min
ALL	5,885 min	5,383 min	502 min

The arithmetic average is 58.85 minutes per baseline load and 53.83 minutes per alternative load. The median comparisons (59.0 and 53.5) are separate distribution measures. Neither method establishes causation without actual source transactions and defensible comparisons.

07 Financial ledger reconciliation

The workbook records fictional billed amounts and three types of direct cost. Independent recomputation of the 100 inputs yields the same financial totals as DASHBOARD and COST MODEL.



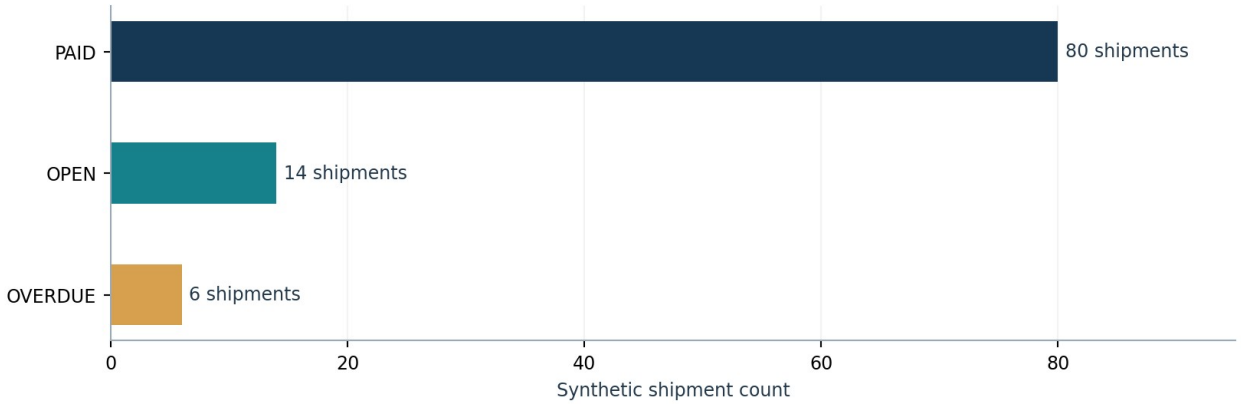
Cost bridge	Illustrative USD
Billed amount	$47,461.52
Material cost	$28,023.36
Freight cost	$5,002.30
Other direct cost	$812.50
Total direct cost	$33,838.16
Calculated direct contribution	$13,623.36
Contribution / billed amount	28.70%

ACCOUNTING CLASSIFICATION

Direct contribution excludes corporate overhead, depreciation, income taxes, interest, financing costs and fixed-cost allocations. Fuel/equipment records are not automatically allocated to direct costs. Do not label these results as net income, EBITDA, cash flow, verified savings or GAAP financial statements.

08 Receivables and equipment calculations

The receivables ledger classifies all 100 synthetic invoices as of the fixed CONTROL!B7 date. The classification, amount totals and paid-invoice lag were inspected against the workbook.



Status	Shipments	Illustrative billed amount
PAID	80	$37,718.24
OPEN	14	$6,705.41
OVERDUE	6	$3,037.87
ALL	100	$47,461.52

The mean time to collect for paid invoices is 15.675 days. This is not Days Sales Outstanding (DSO). Unpaid receivables total $9,743.28, based on the demonstration as-of date. An updated review date changes the classification context.

Equipment ledger

A separate set of 12 synthetic equipment-period records uses productive/idle hours, liters consumed, reported tonnage and fuel rate. Calculations include utilization, fuel intensity and sample fuel cost. Its numbers are not reconciled to TRANSACTIONS fuel inputs. It is incorrect to aggregate or combine them as a single cost pool.

09 Integrity, sensitivity and recalculation findings

The unchanged synthetic input sample contains zero duplicate shipment IDs, zero transaction CHECK flags and zero equipment CHECK flags. These outcomes are directly reflected in QA TESTS; they do not prove that all anomalous inputs would be detected.

Limited test in a temporary copy	Observed behavior	Disposition
Negative shipment quantity in TRANSACTIONS row 7	Row-level data integrity changed to CHECK; aggregate exception count became 1 and QA status FAIL.	PASS (INTERNAL)
Negative equipment fuel in EQUIPMENT LOG row 7	Equipment QA changed to CHECK; aggregate equipment exception count became 1 and QA status FAIL.	PASS (INTERNAL)
Duplicate shipment ID in row 8	Both affected rows flagged DUPLICATE; QA duplicate count rose from 0 to 2; overall QA changed PASS → FAIL.	PASS (INTERNAL)
Modified baseline invoice-reference match	Changing matched invoice reference on row 8 lowered matching count 85 → 84 and dashboard match rate 85% → 84%; test status changed to FAIL.	PASS (INTERNAL)

CORRECTIVE TEST RESULT — INTERNAL PASS

The original duplicate-ID summary used a COUNTIF array operation inside SUMPRODUCT that did not respond to controlled data edits in the workbook evaluation environment. Version 1.1 uses a transparent row-level uniqueness formula in TRANSACTIONS column AM and counts flagged rows in QA TESTS. Controlled tests of duplicate IDs, mismatched ticket reference, negative shipment quantity and negative equipment fuel now propagated to the expected quality-control outputs. The corrected .xlsx was reimported and the duplicate case retested successfully. These internal tests do not replace a full native Microsoft Excel regression or external review.

Additional checks not covered by the 13 reference tests

- Ticket and invoice uniqueness across different shipments, not only duplicate shipment IDs.
- Authorization to issue invoices, invoice-value matching to accounting records and evidence of receipt.
- Invalid dates, timezone changes, exceptional service conditions and cross-ledger fuel allocation.
- Formula coverage after inserting, deleting or sorting rows; recovery after file corruption or accidental input over formula cells.

10 Known limitations and risk register

Risk	Why it matters	Required control
Synthetic evidence	Fabricated numbers have no business-effect inference.	State synthetic at every public use.
Alternative scenario	Constructed improvements are not observed pre/post.	Replace with controlled field study.
Reference-specific QA	Hardcoded expected values fit one original cohort.	Build adaptable unit tests for live data.
Limited input constraints	A PASS cannot prove completeness, identity or source authenticity.	Introduce independent source audit and anomaly tests.
Unlinked equipment costs	Potential omission/double counting if misused.	Source-based equipment allocation policy.
Limited row coverage	Formula blocks end at transaction row 206 and equipment row 46.	Capacity regression before scaling.
Static as-of date	Receivable classification becomes stale over time.	Controlled date update and freeze.
No access-control guarantee	Excel is a file, not a governed database.	Protect source records and access rights.
No independent assessment	Present report is internally produced.	Third-party review and signed findings.

Immediate risk decision

RELEASE CLASSIFICATION

Suitable for demonstration, training, formula inspection and discussion with potential reviewers. Not suitable for live production data, audit assurance, investment decisions, claimed operational savings or substantiation of U.S. national importance without additional independent evidence.

11 Proposed real-company validation protocol

This is a prospective protocol consistent with the Professional Plan; it has not yet been conducted. Actual data collection requires written permission, confidentiality controls and, where the work occurs in the United States, appropriate immigration and professional authorization.

Gate	Minimum evidence	Acceptance decision
G1 Authorized access	Signed scope, data controller, purpose, confidentiality and permitted retention.	No source access without authorization.
G2 Baseline mapping	Preserved ticket, billing, dispatch and equipment source extracts; defined units.	Reject unexplained field gaps.
G3 Pretest baseline	At least 8 weeks of usable baseline where feasible (§16.1).	Document exceptions and mix changes.
G4 Controlled action	Named business process owner, approved change date, safety review.	No unapproved changes.
G5 Follow-up	At least 8 weeks under comparable definitions where feasible.	Do not imply causation without defensible design.
G6 External verification	Independent professional reproduces measures; records adverse findings.	Report null results and limitations.

Minimum output package for a defensible pilot

- Source-to-field mapping, dated data dictionary, chain-of-custody record and permission letter.
- Versioned Excel file, read-only source extracts, frozen metric definitions and automated test records.
- Baseline/post-period reconciliation, sample-size disclosure, source completeness, confounder log and unfavorable outcomes.
- Reviewer's dated technical observations, reproducibility notes, qualifications and authorized release permission.

12 Relevance to EB-2 NIW evidence

A project-specific technical artifact may substantiate preparation and capability to execute a defined proposed endeavor. The report should be used for that narrow claim and accompanied by the actual workbook, version documentation and independently verifiable record of authorship.

Potential proposition	Support furnished by v1.1	Still missing
A working calculation framework exists	Yes — selected numerical and formula functions demonstrated.	Broader reliability/field suitability.
The applicant is positioned to develop it	Partly — artifact and documented technical specification.	Attribution, permitted work samples, firsthand corroboration.
U.S. small firms need practical controls	Not established by sample outputs.	Separate authentic federal/sector sources and user research.
The framework has national importance	No — a technical demonstration is insufficient.	Genuine wider implications and independent relevance.
U.S. companies have used the method	No — there is no pilot or customer.	Actual authorized use, reports and independent adoption.

USE IN AN IMMIGRATION RECORD

Do not present this internal report as government approval, a neutral expert opinion, evidence of U.S. deployment or independent validation. Any future reviewer must be free to disagree, document defects and give an authentic first-hand technical assessment.

Recommended evidence sequence

1. Preserve the corrected v1.1 workbook and checksum. 2. Complete native Microsoft Excel recalculation and wider adverse-input regression tests. 3. Document authorship and version history. 4. Obtain competent independent technical review. 5. Pursue authorized industry feedback and, if possible, a properly scoped field pilot.

Appendix A Reproduction record and review sign-off

Core reconstruction recipe

For i = 1..100: base_minutes = 40 + 4 × (i mod 9) + (i mod 7); alternative_minutes = base_minutes − ((7 × i) mod 11); baseline matched when i mod 13 ∉ {0,1}; alternative matched when i mod 31 ≠ 0. Summaries must reproduce 85/97 matches; 5,885/5,383 minutes; medians 59/53.5; 502 artificial difference.

Verification mapping

Result	Workbook reference	Status
13 reference tests	QA TESTS!A7:E19; summary A22:D22	Verified on unchanged file.
100 transaction records	TRANSACTIONS!A7:AL106	Source and computed values inspected.
12 equipment records	EQUIPMENT LOG!A7:R18	Input and direct QA status inspected.
Financial totals	DASHBOARD!B19:B21; COST MODEL!A7:K12	Independent input recomputation matched.
Dispatch outcomes	DISPATCH ANALYSIS!A7:H12	Scenario arithmetic matched.
Receivables	RECEIVABLES!A7:D16	Counts and amounts reconciled.

Independent external review — not completed

Reviewer name / credentials: _____

Organization / contact: _____

Verification environment and software build: _____

Workbook SHA-256 verified: [ ] Yes [ ] No

Adverse-input tests independently completed: [ ] Yes [ ] No

Reviewer findings attached: [ ] Yes [ ] No

Signature / date: _____

FINAL TECHNICAL DISPOSITION

Internal sample validation complete with explicit limitations; external technical certification, live operational validation and field-level effect remain unperformed.

Source specification: Ricardo Lopes Wiest, Professional Plan and Integrated Evidence Dossier, PCSPF, November 2026, §§4, 15, 16, 22–23. Companion User & Implementation Guide v1.0. This report adds no independent government or market-performance claim.